

Christian Abou-Ezzi

Biomedical Engineer
Boston University

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Gesture Sensor Demonstrator

Christian Abou Ezzi, Eve Cruanes, Paula Garcia
Gasco Solano, Devrim Yesilyurt

Instructor: Christine Mulvey

Boston University

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SUMMARY

Over the course of 12 weeks, a hygiene dispenser was created using the APDS9960 gesture sensor as the foundation of the design. Our client asked us to produce a product that would demonstrate the sensor. The requirements were to include an actuation mechanism including at least three gestures. The device aims to use an APDS9960 gesture sensor to pick up a specific gesture and dispense a desired product based on the gesture. The products that were dispensed are pads, tampons, and toilet paper.

In order to create the hygiene dispenser, a series of designs, models, and tests occurred to ensure the best possible and efficient design. Our best design consisted of stepper motors to actuate the dispensing process. The gesture sensor and the stepper motors were integrated using a simple circuit yet very complex code. The product does require a 5 volt 2 amp power supply meaning an outlet must be in range to use the product. Having an outlet may be troublesome for the users resulting in limited areas of usage for our product in bathrooms. A rocker switch connected to the power supply allows for the user to turn the device on and off.

We evaluated the performance of the dispenser through a series of tests noticing that not all the objectives were met. Ideally we hoped to reach a 95% gesture reading accuracy but unfortunately only got about an 80% due to the inconsistency of the gesture sensor. However, our other objectives and all the constraints given by the client were met, indicating a successful product.

The dispenser ensures optimal hygiene by removing direct contact with the dispenser. It hangs by two ropes on a wall allowing for easy access for all heights and ages. There are three distinct gestures that are required to actuate the dispenser, which consists of a swipe left, a swipe right, and a downward swipe which will initiate the proximity sensor. The gestures are kept simple to limit the complexity of using the product and to ensure an efficient way to grab the desired hygiene product. The product ensures a sanitary, efficient, and fun way to start a morning.

INTRODUCTION

Acquiring sanitary hygiene products in an efficient and touchless way is essential for an individuals routine and well-being. Especially if this product was placed in a public bathroom, sanitization is a priority. The dispenser is designed to take up limited space in the bathrooms, hallways, public areas, or wherever the user desires to put it. The purpose of this device is to sense a gesture through a sensor that dispenses different hygiene products, including sanitary pads, tampons and toilet paper, to individuals of all ages. By promoting healthier habits and routines, the dispenser contributes to overall well-being and hygiene maintenance.

KEY OBJECTIVES AND METRICS

There are a variety of objectives that begin with the sensor recognition accuracy. Using table 1 as a reference, there should be a 95% accuracy for the sensor accuracy picking up the right signals. We understand that when younger children use this dispenser, their hand signals may not be as clear to notice which way their hand moves, which can cause some detection issues and dispense the wrong products or cause no reading at all. Another key objective is the devices' response time. If the goal of the project is to create a more efficient way to obtain hygiene products, the gesture reading to the accusation part should occur all within three seconds, which the sensor does achieve. In terms of weight, because the dispenser will contain a variety of products, it must be light enough to transfer it to desired rooms and locations.

Also, since the box will be suspended in the air, we wanted to make sure that the box was light enough so that the device would not fall and potentially break apart. Referring to table 1, the dispenser must be the size of a shoe box and weighing less than five kilograms for easy portability. To limit the weight, the dispenser contains a limited amount of products with six tampons, 3 pads, and a roll of toilet paper.

CONSTRAINTS

For the constraints of our project, there must be an actuation part to the project including the gesture sensor, which in this case is the spinning of the motors. We also needed to include at least three gestures to interact with the gesture sensor. The project must also cost less than 200 total US dollars. In terms of size and portability, the size of our product needed to be at most the size of a shoebox and measure less than 5 kg. This ensures that the user is able to carry the product and place it wherever they like. For older users of the product, a heavier weight would also make it very difficult for them to carry or suspend it.

ELEMENTS OF THE DESIGN

The project consists of three motors, three green LEDs, an arduino, a gesture sensor, a rocker switch, a power source, and a variety of hygiene products as well as 3D prints. Referring to table 2, the type of motors used are the 28byj-48 stepper motor. All our components in our design work efficiently, leading to our desired functionality of the product. For example, the motors can rotate a full 360 degrees, contain a lot of torque, and require very little power. The motors will have three components to work with, one with a chamber for the tampons, a roller for the toilet paper, and the chamber for the pads. While all of these product components have weight, the products have weight too. According to table 4, the motors are valid to use in the dispenser as they can spin up to about 700 grams, which is beyond 313 grams, the heaviest component that contains the tampons too.

The APDS 9960 gesture sensor is the basis this project is designed around. It can detect 5 distinct gestures, a left to right, right to left, down to up, up to down, and a proximity gesture. The three distinct gestures being used are the left to right, signaling the center motor to move, dispensing the tampons, the right to left to signal the left motor to move, dispensing the pads, and the proximity sensor initialized with a downward swipe, requiring the right motor to actuate, dispensing the toilet paper. As seen in Figure 1 and table 5, the APDS 9960 gesture sensor is a viable option. The proximity gestures are meant to be detected from at most 78mm in our testing however the product claims it can detect up to 100mm. The swiping gestures were picked up from a variety of distances with a 100% reading rate up to 5 inches, and the reading rate deteriorating at later inches such as 7 to 8 inches. The proximity, of course, will not be read from a far distance. Its minimum proximity is about 2 mm away from the sensor while the proximity at a max had 100 percent readings from about 77 mm away from the sensor. By keeping the user's hand at a close proximity, the desired product will be dispensed correctly.

The three green LEDs will stay lit indicating that the components are full, but after a product has been dispensed, the green LED on the designated side of a product will turn off. The LEDs reset by turning on and off the dispenser with the switch. In terms of the power source, the arduino will power all of the components inside the dispenser including the motors, LEDs, and the gesture sensor. The power source is a 5 volt 2 amp plugged into a wall that is monitored by a rocker switch. For the housing of the project, the box is made of black acrylic, 3/16th thickness, which is meant to be both visually appealing, and strong enough to avoid breaking of the dispenser when suspended in the air.

DESCRIPTION OF THE FINAL PRODUCT

The final product housing is made of black acrylic, with each wall made by using a laser cutter and attached together using acrylic glue. A switch is on the front of the housing to turn on or off the device while it is plugged into an outlet by a power source. There are three stepper motors within the housing that move three different mechanisms. The first motor, for the pads, is attached to a rod which is attached to a rope connected to the pad chamber, which is split into 8 compartments. As the motor rotates, the pad chamber moves forward, aligning the compartments with a hole under, allowing the pads to fall out the bottom of the enclosure for the user to grab. The second motor spins a cylinder that consists of 7 holes, one with nothing in it so that the barrel can start somewhere without dispensing immediately. The rest of the holes are filled with tampons. As a right gesture is sensed, the motor for this barrel actuates and spins the cylinder, having the tampon drop through the hole and dispensing it. For the last motor, if a downward motion is sensed and a user places their hand to the sensor, the last motor spins a rod holding toilet paper. There are three green LEDs on the front of the housing indicating when the products are full or not. When a product is empty from a chamber, the LED will turn off and can be reset by flipping the switch. The gesture sensor is also on the front of the housing to make the gesture easily readable.

EVALUATION OF RESULTS

Many metrics were met, but some were not. For the gesture recognition, the correct gesture was only correctly read about 80 percent of the time. Causes for the imperfection can be deducted from putting the hand to the gesture sensor itself. Sometimes when the hand is brought close, a gesture can be read, sometimes the wrong one, dispensing the wrong product. This cannot be resolved as this is a problem with the sensing of the gesture sensor itself. Another metric not met was the refill frequency. Due to inadequate designing, the pad chamber can only hold 3 pads instead of the 8 requiring a frequent refill. Yet the tampon chamber can hold as many as needed. However, everything else has been met. The product weighs less than 5 kilograms as attended for easy portability, and the availability of the product to not fall apart when it is held up. The gestures can all be read within three seconds allowing for an efficient response and dispense time. The product has a low power consumption as seen in table 6. The product also costs less than 200 dollars, which clears the metric as seen in table 2.

LESSONS LEARNED

Over the course of designing our product, we learned how to navigate through difficulties, code for the actuation parts of the project, and learned how to work together in sharing ideas, turning them down, and building off of other ones. The wiring aspect of the project was one of the most difficult as many of us just started taking circuits, or knew nothing at all about how circuits work. Using knowledge from past classes such as EK 131, a working circuit was built successfully to integrate the gesture sensor and stepper motors to work together. After figuring out the wiring, coding everything together was another challenge faced. Transferring the signals to the motors and working the LEDs all at the same time was very time consuming. We tried adding buttons, but ended up learning how some materials cannot work with others. We learned that buttons and the gesture sensor cannot work together. The signals of both interfere with each other, not allowing us to put the buttons in our project. Lastly, we learned how to create designs with a team. The hardest mechanism to create was definitely the pads chamber. About three weeks were put into creating the chamber whether it came with testing the designs we made, failure of the

designs during production, or just realizing none of the options would work in general. However, in the end, we all pitched in ideas and combined them to make the pad chamber work.

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Table 1: Objectives and metrics of the project

Objectives	Metrics
Sensor recognition Accuracy	> 95% responding to the gestures correctly
Weight	No more than 5 kilograms
Response time	Must be able to read the gesture and start performing an action within 3 seconds
Portable	Less than 5 kilograms, about the size of a shoe box.
Refill Frequency	Contains 6 tampons, 8 pads, a roll of toilet paper
Power consumption	10 watts of power, cheap consumption
Cost	Less than \$200

Table 2: Costs of the project

Component	Quantity	Cost per component	Total cost
APDS 9960 gesture sensor	1	\$13.95	\$13.95
28byj-48 stepper motor	3	\$4.95	\$14.85
Arduino	1	\$27.6	\$27.6
Red wires	9	0	0
Black Wires	9	0	0
Green Wires	4	0	0
Purple Wires	4	0	0
White Wires	4	0	0
Brown Wires	4	0	0
Rocker switch	1	\$0.67	\$0.67
Green LED	3	\$0.56	\$1.68
220 Ohms Resistor	3	0	0
5 volt 2 amp power source	1	\$7.95	\$7.95
3D print cylinder	1	\$9.00	\$9.00
3D print paper roll holder	1	\$2.75	\$2.75
Pad Chamber	1	0	0
Tampons	6	0	0
Toilet paper roll	1	0	0
Pads	3	0	0
Rocker Switch	1	\$1.15	\$1.15
Housing (Acrylic 12x24)	4	\$17.87	\$71.48

Eyebolts (4 pack)	1	\$15.99	\$15.99
Hinges	2	\$2.83	\$5.66
TOTAL COST			\$172.74

Table 3: Morph chart of the hygiene product dispenser

Design Element	Option 1	Option 2	Option 3
Gesture Recognition	APDS 9960 Gesture sensor	Infrared sensor	-
Motors	Stepper motor	Servo motors	DC motors
User interface	Green LED	RGB LED	Display panel
Power supply	5V 2A power supply	Battery-power	-
Housing material	Acrylic	Plastic	Metal
Product dispensing	Candy	Hygiene products	-

Table 4: Results from the motor performance analysis based on the supported weight.

Weight (Grams)	Does it Spin?
<100	Yes
100	Yes
163	Yes
263	Yes
313	Yes
>700	No

Table 5: Results from the functionality of the gesture sensor based on the maximum and minimum proximity and the distances (mm)

Test #	Min. Proximity and Distance (mm)	Max. Prox and Distance (mm)
Test 1	0 at 2mm	255 at 77 mm
Test 2	0 at 2mm	255 at 78 mm
Test 3	0 at 2mm	255 at 76 mm

Table 6: Power Budget Calculations

Component	voltage	current	Power = voltage * Current
Arduino	5	.05A	.25 W
28BYJ-48 Stepper Motors(3)	5	.2 A each	3 W total
Green LED (3)	5	.02 A	.3 W total
APDS 9960 gesture sensor	5	.005 A	.025 W

Gesture Recognition Success Rate At Varying Distances Away

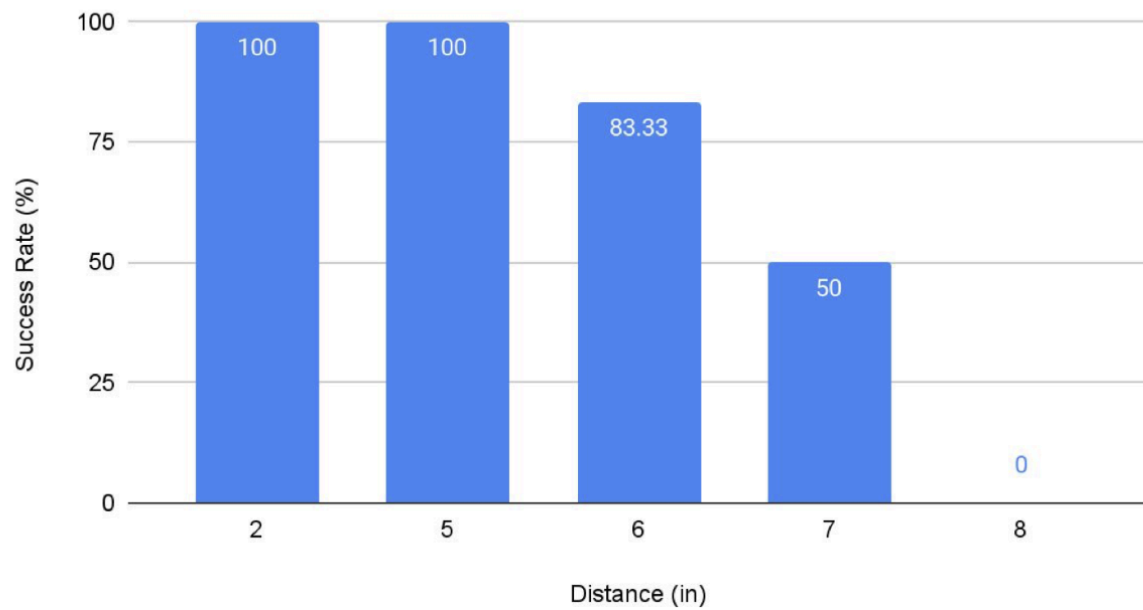


Figure 1: Success rate (%) of the gesture sensor as a function of distance (in)

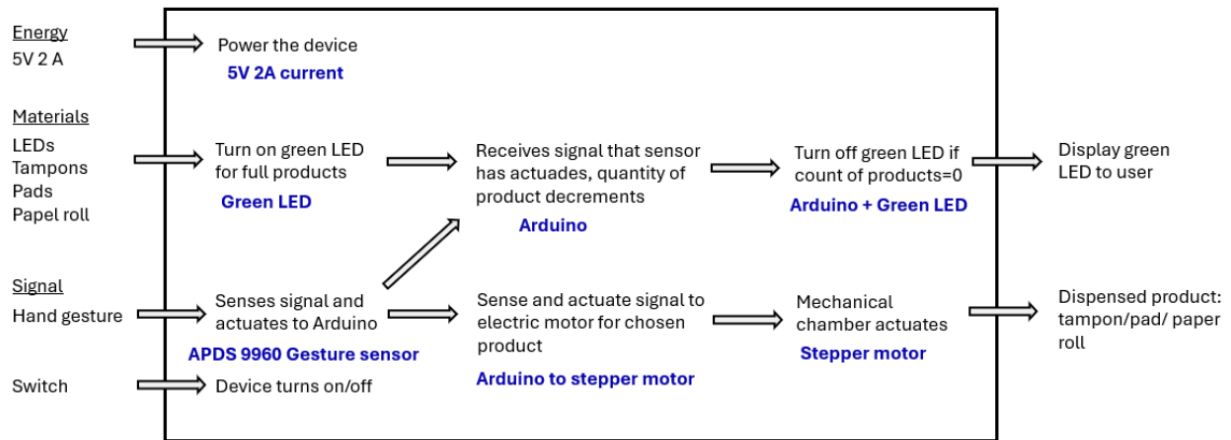


Figure 2: Glass box of the hygiene product dispenser

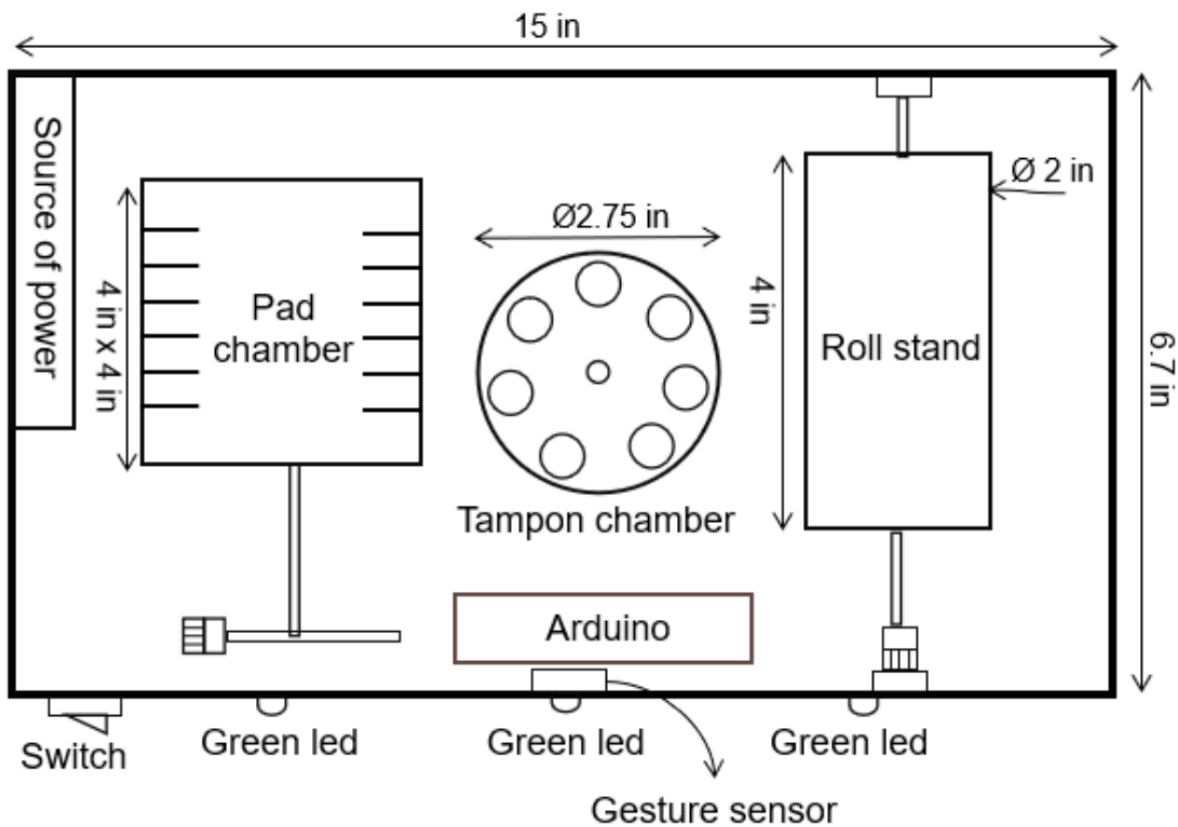


Figure 3: Top view sketch of the hygiene product dispenser.

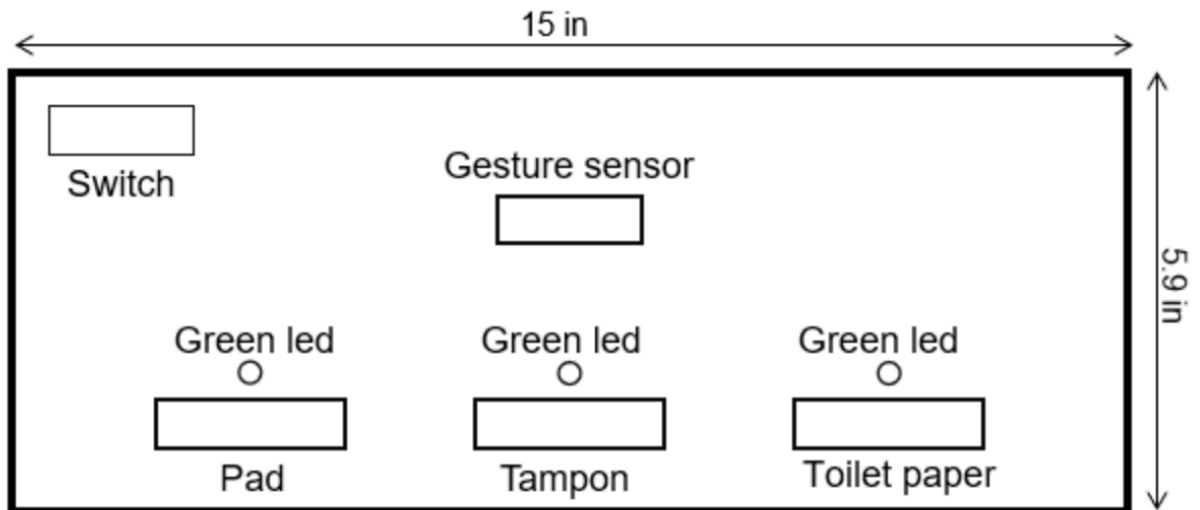


Figure 4: Front view sketch of the hygiene product dispenser

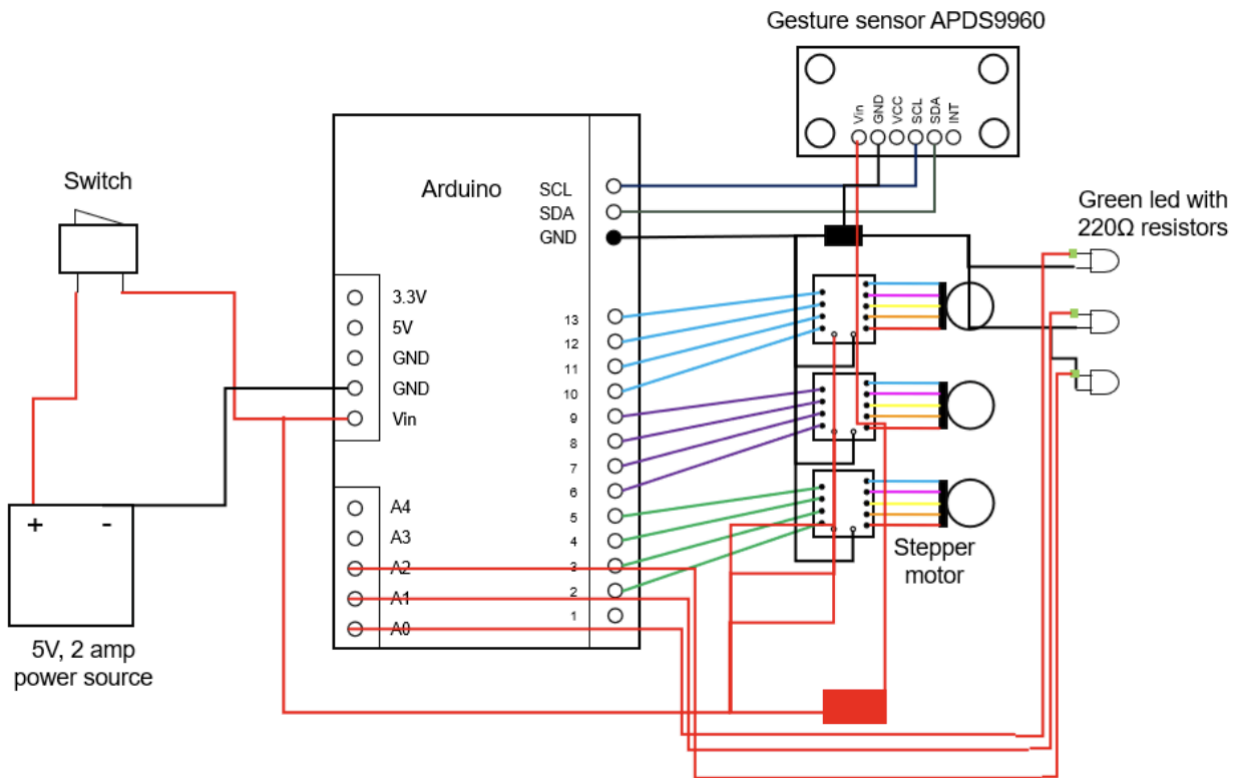


Figure 5: Wiring diagram of the hygiene product dispenser

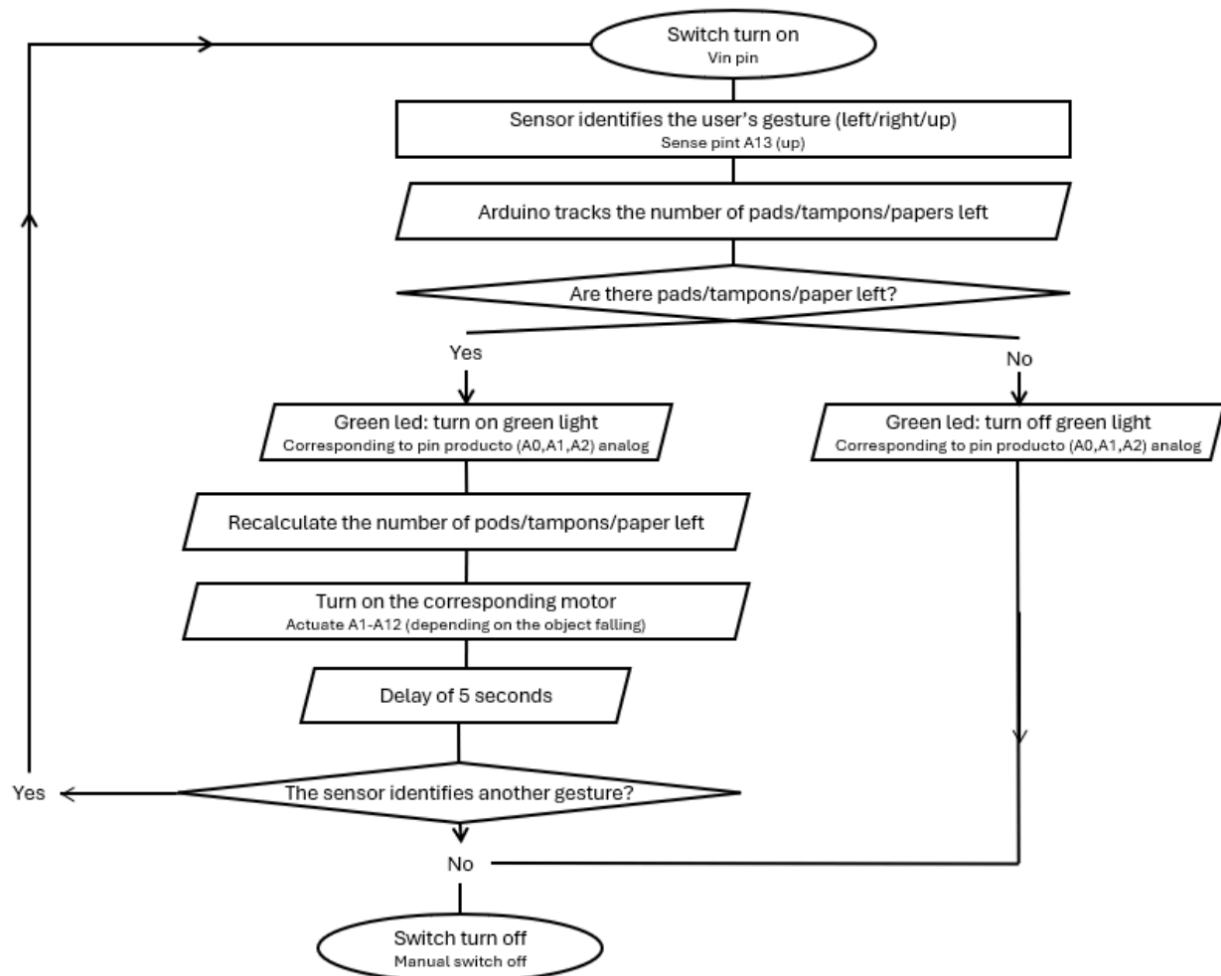


Figure 6: Code flow for the hygiene product dispenser.

Figure 7: PCC Chart

Goals	Accuracy	Weight	Response time	Portable	Refill frequency	Power consumption	Cost
Accuracy	—	1	1	1	1	1	1
Weight	0	—	1	0	0	1	0
Response time	0	1	—	1	1	1	1
Portable	0	1	0	—	0	0	0
Refill Frequency	0	1	0	1	—	0	1
Power consumption	0	1	0	0	0	—	0
Cost	0	1	0	1	1	1	—

Scores:

Accuracy: 6

Weight: 2

Response Time: 5

Portable: 1

Refill Frequency: 3

Power Consumption: 1

Cost: 4

Final Design Report

Arnouv Nayana, Christian Abou-Ezzi, Max Lopez

Instructor: Scott Bunch

Boston University

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Introduction

The goal of this design project was to design a truss that maximized one or more parameters, while staying within the given restrictions. When designing all iterations of our truss, but especially for the final design, the primary goal was to learn how to create and have the largest maximum load possible. Since a budget is given, we believe that underutilizing it is not going to help anyone, but making the structure stronger will allow it to be safer and more useful. If the truss is treated like a bridge, then making sure the bridge can support as much mass as possible is of the utmost importance in order for it to be as safe as possible. This thinking guided us to our final design. Additionally, we inherently were able to maximize our load-to-cost ratio since we hardly increased our price as we developed the design. While this was not a particular “goal”, it ended up becoming our second strongest consideration. In tandem with this, we did not aim to minimize uncertainty, instead seeking to maximize the predicted strength as much as possible. Even within the uncertainty range, the maximum load should have a high value that we believe would be viable for the final design.

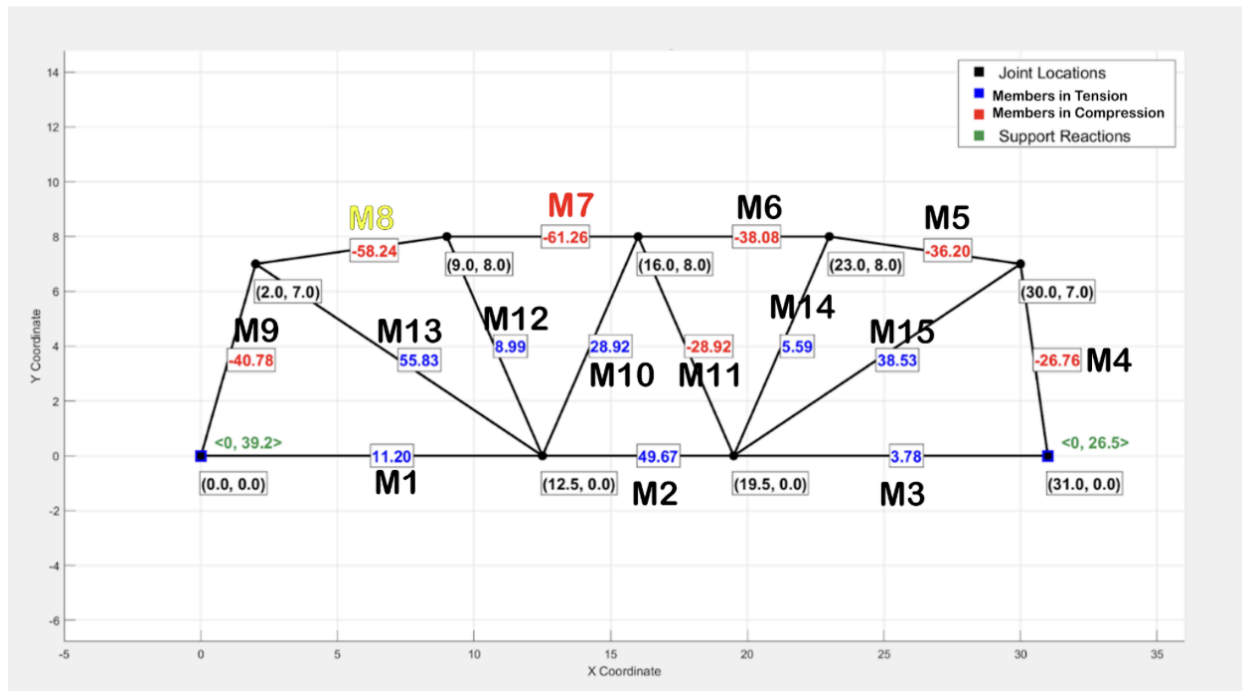
Procedure

Our procedure as outlined in the preliminary design report has gone unchanged. In short, this entailed tweaking parameters within our Matlab program one at a time to see the effect each one has on the results. Eventually, those tweaks became smaller and smaller as a design was honed in on that maximized our goals while not significantly impacting other parameters.

Analysis

The design between the preliminary and final iterations are similar in a general sense, but have a few key differences based on further understanding. We realized that evening out the design would improve it in this case since it would take some strain off a few members. We also learned during our tweaks that making it more rounded along the top improved the design, so we made sure our final iteration accounted for this. There were attempts to create an even more rounded top in some further iterations that were more effective, however those had members that were outside of the 7-14 inch range allowed, and further adjustments to remedy this proved to result in similar issues. The analysis that was used to determine this was identical, although it was applied in another way in this phase to determine uncertainty values for each member in compression. Within our code analysis, to determine the failure weight, we already have been calculating a proportionality constant (R) for each member, which is used in the equation $Length = R * W_{fail}$. This formula was used to determine each member's strength at the failure weight. We additionally used $W = 3054.789 * L^{-2.009}$ to determine the buckling strength of each member based on the member's length. These in turn let us determine if any members were within the uncertainty range of being the member that will fail instead of the critical member.

Results



Final Truss Design - Group "Number 15"

This image was generated using Matlab

The critical member of this truss is Member 7 highlighted in red above. It has a length of 7 inches and a buckling strength of 61.26oz. The maximum theoretical load of this truss is 65.698 oz, it will cost \$223.66, and the load to cost ratio is 0.2937 oz/\$. When accounting for the uncertainty range for the internal forces at the given load, Member 8 is within the failure range, and will potentially fail first instead of Member 7. It is highlighted in yellow above.

Results Table

Member #	Length (inches)	Tension/Compression	Internal Force @ 65.698oz (oz) Uncertainty (+/- 1.36)	Buckling Strength (oz)
M1	12.5	Tension	11.202	N/A
M2	7	Tension	49.671	N/A
M3	11.5	Tension	3.784	N/A
M4	7.1	Compression	26.760 25.400-28.120	59.539
M5	7.1	Compression	36.205 34.845-37.565	59.539
M6	7	Compression	38.081 36.721-39.441	61.26
M7	7	Compression	61.261 59.901-62.621	61.26 Critical Member
M8	7.1	Compression	58.243 56.883-59.603	59.539 Potentially will fail
M9	7.3	Compression	40.776 39.416-42.136	56.307
M10	8.7	Tension	28.916	N/A
M11	8.7	Compression	28.916 27.556-30.276	39.581
M12	8.7	Tension	8.991	N/A
M13	12.6	Tension	55.832	N/A
M14	8.7	Tension	5.589	N/A
M15	12.6	Tension	38.527	N/A

Discussion & Conclusion

When optimizing the final design of the truss, as mentioned above, we particularly ended up focusing on evening out the layout and rounding out the top. We started with re-adjusting the placement of each member to make it more even. This resulted in us adding two additional members, which further strengthened the truss by taking the strain off of other members. After

this, we experimented with the top part of the design, since this was the least important to maintain for our overall structure. As a result, we discovered the benefit of the rounded top and added it to the design. The most beneficial aspect of the project was the process of creating new designs to try to improve our maximum load. From this, we were able to naturally see what works well and what does not work as well as we expected. This is something that is easier to learn in an experimental context, since in a theoretical context it is difficult to actually visualize a design that would be optimal just from equations, so being able to see and tweak a design is ideal. In a future run of this project, we would create more unique and varied designs after the preliminary design, still using our knowledge from that stage, but applying it in a wider variety of base shapes and structures. Additionally, with additional time, we would further pursue the more rounded design that we were unable to get working within the scope of the project, since we believe that would be even more successful.

Christian Abou-Ezzi

Design Process of Room Temperature Monitor

1. Summary

The prototype created is a room temperature monitor that can read the temperature of the room in both celsius and fahrenheit and display it on an LCD. There is a song always playing throughout the reading, however, when the temperature is outside the specified range, 68-75 fahrenheit, a red LED flashes and the buzzer makes a beeping noise. Creating the prototype began with mounting the arduino board, the battery holder onto the mounting plate inside the enclosure, and the LCD on the lid. Next, many wires were soldered together and color coded. The soldered wires were covered by heat shrink to prevent the wires from being exposed. The buzzer and green LED were placed into the wall of the enclosure while the temperature sensor needed double sided tape to be put onto the wall, and the red LED and a piece of plastic slipped under it so it would not fall out. To power the circuit, wires were connected to the arduino, LCD, the grounds, and 5V connection. A switch was also placed into the wall of the enclosure connected to the battery and arduino to power the circuit from a switch. After the circuit was built, a code was uploaded to the arduino. The green LED turns on when the monitor is turned on. When the temperature of the surround exceeds 75 degrees or is below 68, the red LED blinks and the buzzer makes a sound, all while a Harry Potter song plays. The temperature in Fahrenheit and celsius will then be displayed on the LCD and refreshes whenever the song ends.

2. Introduction

The device is useful as it records and displays the user what the temperature of the surrounding is. This is useful for people who need a room set to a certain temperature for a pet or food reasons without having to constantly look at a thermometer to read the temperature. One aspect of temperature control can relate to having a reptile-like pet such as a snake or lizard. These animals are cold blooded and are required to have a certain temperature in their surroundings to maintain homeostasis. This temperature monitor can help read the temperature of the surrounding environment while playing a soothing melody. If the monitor beeps, the temperature is outside the range that the room should be at. For this reason, the reptile owner can change the temperature of the room to the desired limit. This can determine if the heat needs to

be turned on if it is too cold, or for the air conditioning to be put on if the temperature is too warm. When it is within the desired range, there will be no beeps before the playing of the song. To build this prototype, the website used was CAD to help design the lengths needed and the components used for the device. CAD was also used to design the battery holder which was later printed with the same exact measurements used in the CAD drawings to fit the 9 volt battery into the enclosure and into the battery holder. A temperature sensor also was put in after the battery holder to read the temperature onto the LCD. Putting everything together, I learned how to build and work a circuit. Building the circuit required a lot of soldering, heat shrinking, and combining wires to an arduino board to work with a code that was made. When something went wrong in the building process, I was able to learn how to use a multimeter to track where the problem stems from. This report will outline the process and result of building the prototype and the details of the components used too.

3. Design elements

- a. The components used were: an ABS enclosure, an Arduino Uno Board, a two-way switch, an Alphanumeric 12C LCD 16X2, a buzzer, a temperature sensor(TMP36), a red and a green LED light, a 9 volt battery, one brown wire, two red wires, 7 black wires, one yellow wire, 2 white wires, 4 blue wires, two resistors one that is 220 ohms and the other that is 1000 ohms, two cutouts of a green board, two pieces of double sided tape, 8 black nuts, 4 black screws, and a USB cable. The solid core jumper wires and the 22 AWG solid wires refer to the different color wires used.
- b. Precision measurements (Figure 1)
- c. CAD drawings (Figure 2)
- d. Prototype pictures (Figure 3)
- e. An Arduino has a physical circuit board and a software that can be used to take in inputs from a code and perform tasks of utilities wired to the arduino properly. Arduino board is used for control and converting analog signals to digital signals. In my circuit, the Arduino is meant to power my buzzer, LEDs, LCD, and temperature sensor. This is achievable by connecting the black wire(negative) to the grind outlet on the arduino and the red to the VIN. This will be the source of power for everything else that is connected to the arduino. The LCD is connected to the arduino with A0, A1, A2. A0 is to the

temperature sensor to read the temperature, A4 to SDA which would be the data signal, and A5 to the SCL which is the clock signal. These combined will read the temperature and output data to the LCD powered by the arduino that is connected to a battery. The Arduino also has taken my code to command what operation functions such as when an LED flashes, when to read the temperature, and what to display on the screen.

- f. To connect the black and blue wires together as seen in figure 4, soldering the wires occurred. Resistors were also soldered onto the LEDs and the rest of the wires were soldered to the sensor and buzzer. To cover the soldering and to prevent exposed wires, heat shrink tubes were used to cover the connected black wires and the connected blue wires. They were also used to cover the resistors so the wires are not showing. A battery snap was used to connect the battery to the switch and arduino. Wires were also soldered to the LCD to the other components in the electrical circuit. (wiring diagram in Figure 4). The pins used are A0, A4, A5, 5V, the two grounds, Vin, digital pin 5 and 3. A0 has an orange wire connecting to the middle stem of the temperature sensor for power, The black connects to all the other black wires, and the blue is connected along with the other bundle of blue wires. A4 is connected to the SDA of the LCD which is the signal data for the LCD. A5 is connected to SCL of the LCD, the signal clock for the LCD. 5V is connected to the bundle of blue wires to power the blue wires. The two grounds connect one to the battery and the other to the bundle of black wires. This will power everything from the battery. Vin is connected to the switch which is connected to the battery to be turned on by the switch. Pin 5 is connected to the positive side of the buzzer to power it and pin 3 is connected to the positive side of the red LED and resistor to power it.
- g. For the data connections, 26 AWG wires are used because they are smaller and are better for signal connections. The 26 AWG wires are 361 mA, which is enough to transfer the data. The 22 AWG wires are used for power. The max current those wires can take are 920 mA and the max current drawn in the circuit is less than 100 mA. Wires with a smaller gauge can carry more power than larger gauge wires making them a better fit for power. The green LED uses a resistor of 1000 Ohms while the red uses 220 Ohms. The operating current of the LEDs was 80.6 mA, which was found using a multimeter on the switch inside the enclosure.

- h. The Arduino board accepts voltages of 6-20 volts. Because of this, the 9 volt battery was used since it falls within this range. To power the circuit, the switch was connected to the battery to allow it to turn on and off the circuit by blocking or unblocking the flow of the current.
- i. Arduino code (Figure 5)
- j. The power supply has a voltage of 9 volts drawing power from the 9 volt battery. However, the voltage output by the arduino can only reach 5 volts because that is the max output voltage. The current measured throughout the circuit was 80.6 mA recorded by a multimeter. The temperature ranged from the spec sheet and design is 68-75 degrees fahrenheit. The operating time of a 9 volt battery is 7-7.5 hours.

4. Evaluation of Results

The prototype copies the wire diagram as seen in figure 4 exactly as follows. The following prototype is also built exactly as shown in the CAD drawings. Every component functions as explained with the temperature sensor reading the temperature, the LCD displaying the temperature, the buzzer plays a soft tune while it reads but buzzes when outside the temperature range of 68-75 degrees, the red light flashes whenever the temperature is outside the desired temperature, and the code executes as exactly as intended to do so. Compared to standard thermometers, this prototype is battery operated and requires a code to function. It also plays a tune and beeps when outside the desired temperature range. Thermometers on the other hand do not make a sound and are also not powered by energy, rather tell the temperature through the fluctuation of the mercury's height in the thermometer container. Some limitations that could apply is the battery life as after about every 7 hours the battery must change. The LCD is also not very bright and is difficult to read, so a bigger LCD would be desirable. The prototype is also much larger and heavier than a traditional thermometer making it harder to transport the device. While it does have some limitations, lighter material and a smaller box would be a great recommendation to improve the prototype, even a bigger and brighter LCD would be better. While the device does have limitations, the prototype can still record the temperature, easily display it, and even notify the reader that the room temperature is outside the desired temperature range.

5. Supporting Materials (appendix)

Item	Width (mm)	Length (mm)	Height (mm)	Diameter (mm)
ABS Enclosure	118.0	118.0	59.0	n/a
Arduino	52.3	67.5	11.8	n/a
Switch	14.1	20.2	25.2	n/a
LCD	80.3	36.5	9.2	n/a
Buzzer	n/a	n/a	18.1	21.8
Temp Sensor	n/a	n/a	18.6	4.5
LED (R & G)	n/a	n/a	36.8	4.9
Battery holder	36	50	28	N/A
Mounting plate	2.9	95	95	N/A
ABS Enclosure Lid	118	118	3	N/A

Figure 1. The measurements of every component used in the prototype.

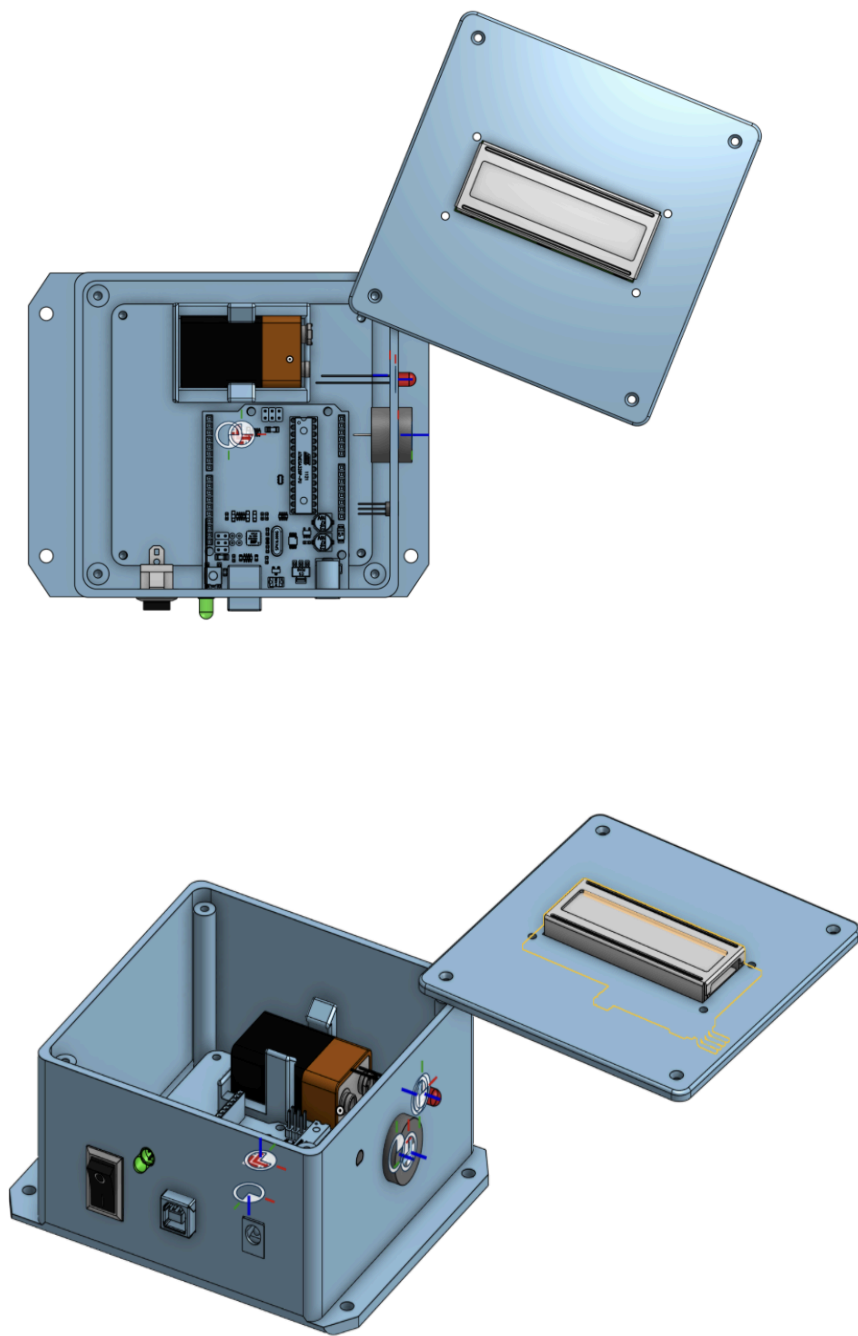


Figure 2. The CAD drawings of the prototype with top and side view.



Figure 3. The side, top, and coverless top view of the actual prototype.

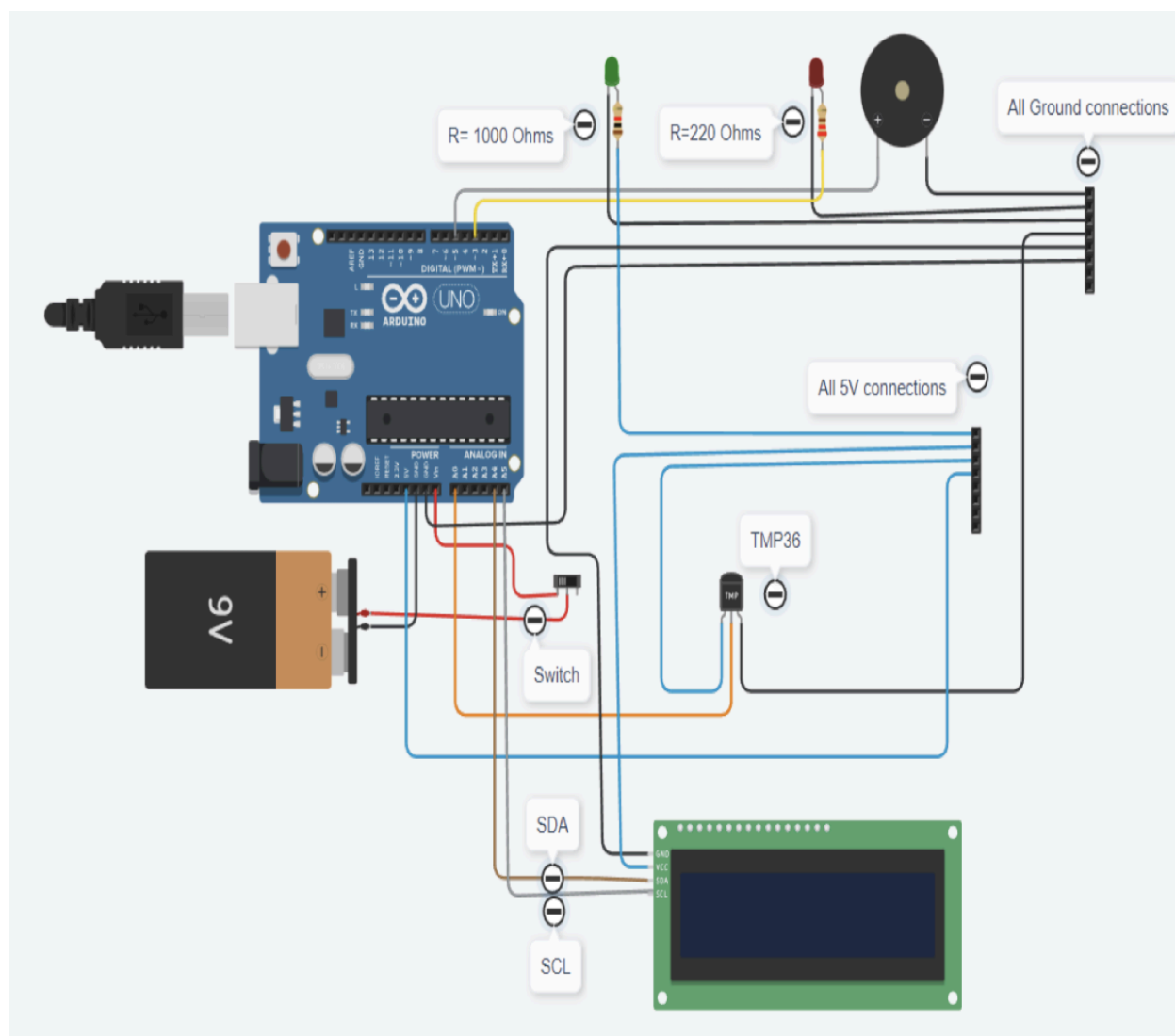


Figure 4. The wiring diagram of how the prototype is wired internally.